

Project 1 — Testing a Claim

Is this observation surprising?

Project 1: Testing a hypothesis from a binomial random process

In this project, you will write a one-page summary of how you used the binomial distribution to draw a conclusion about an infinite process from a finite amount of data that you collected. For the inference from your finite data to the infinite process to be valid, your data collection must satisfy the four conditions of a binomial random process (BRP).

Download [sample one-page write-up — ESP binomial test](#). This is my completed report for an ESP study. Do not copy my process (ESP is a class example you may not use for your own project). Write up your own study following the steps below.

Project workflow

Phase	What
A — Milestone	Plan question, p , H_0/H_a , null graph (no x yet)
B — Full project	Collect $n = 25$, analyze, conclude one-page report

Phase A

Step	Task
A1	Choose a binomial process with $n = 25$ (not a class example). Check all four BRP conditions: (a) binary outcome per trial, (b) independent trials, (c) fixed p per trial, (d) fixed n .
A2	Write your research question with your best guess about the size of p in the infinite process.
A3	Define p in context — what counts as one “success” on a single trial.
A4	State H_0 and H_a with a numeric p ($0 < p < 1$). Remember that you should write H_a by translating your research question to our math notation.
A5	Plot the null binomial distribution in R (<code>dbinom + geom_col</code>) using $n = 25$ and p from H_0 . Do not mark an observed x yet.

Phase B

1. OBSERVE at least $n = 25$ trials. Record number of successes x . Compute $\hat{p} = x/n$.

2. In R, make a descriptive bar or pie chart for your observed counts x and $n - x$ from the n trials.
 3. Mark observed x on the null plot (A5) with a vertical line via `geom_vline(xintercept = ...)`.
 4. Compute the p-value with `pbinom()` (match the direction of H_a). Interpret in context.
 5. Write up a one-page summary using the rubric below (Introduction, Data Collection, Data Analysis, Conclusion).
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Project 1 grading rubric

Total points: 35 (one page summary below) + 5 (.ipynb with all your project code) = 40 points

Format (2 pts)

Your summary should be organized into 4 sections (1 section with 2 subsections) with the following headings:

1. Introduction
2. Data Collection
3. Data Analysis
 - 3a. Descriptive Statistics
 - 3b. Inferential Statistics
4. Conclusion

Style (3 pts)

- Your summary should fit within one side of an 8.5 by 11 inch page. (1 point)
- Your summary should be concise and written in complete sentences with proper grammar. (2 points)

Introduction section (5 pts)

In this section, you should:

1. Briefly describe what question you are trying to answer. (1 point)
2. Briefly describe the motivation you have for trying to answer the question. (1 point)
3. Describe the parameter of interest p in one sentence. (1 point)
4. State the null and alternative hypotheses for your research question using appropriate notation and a value for p . (2 points)

Data collection section (7 pts)

- Describe how, when, and where you collected data in enough detail that someone else could repeat your study. (3 points)
- Explicitly explain why you believe each of the four BRP conditions was satisfied by your method. (4 points)
Explain in the context of your study, not as a generic list of the four conditions.

Descriptive statistics subsection (6 pts)

- Include the total number of trials n , the number of successes x , and the percentage of successes you observed. (3 points)
 - Include a well-labelled pie chart or bar chart of your observed data, made in R (e.g. with `ggplot2` or `barplot`).

(3 points)

Inferential statistics subsection (8 pts)

In this subsection, please include:

- A graph of the binomial distribution for all possible values of X using the p from H_0 . Create this as a column chart in R using `dbinom()` for the bar heights and `geom_col()`. (4 points)
- The p-value: the probability of seeing a new observation of X as extreme as your x if H_0 is true. Compute it in R with `pbinom()`. (2 points)
- A brief explanation of how you calculated the p-value. (1 point)
- A clear indication of where your data x appears on the binomial graph (e.g. a vertical line or label). (1 point)

See the [ESP sample PDF](#) for an example of wording and figures.

Conclusion section (4 pts)

- Write a one-sentence summary of what your p-value means in the context of your research question and data. (2 points) (See the [sample project](#) for an example.)
- Based on the size of this p-value, state what you conclude about your research question. (2 points)